

Lab 17: Packet Capture and Header Analysis by Wireshark (TCP, UDP, IP)

Assignment Date: August 17th, 2026

Deadline: August 26th, 2026

Class: BSc.CSIT 4th Semester, New Summit College

Subject: Computer Network

Objective: The primary goal of this lab is to learn how to capture network packets and analyze their headers using Wireshark, focusing on TCP, UDP, and IP protocols.

Tasks:

1. Install Wireshark
2. Capture live network traffic.
3. Analyze and interpret packet headers (TCP, UDP, and IP)
4. Understand how different protocols behave on the network.

Report Format & Structure (Topics to include for submission):

- INDEX
- Theory
 - a. Wireshark
 - b. TCP, UDP, IP
- Implementation
 - a. Capture the Network Interface of Choice and Filter the traffic
 - b. TCP Header Analysis (include Source Port, Destination Port, Sequence Number, Acknowledgement Number, Flags)
 - c. UDP Header Analysis (include Source Port, Destination Port, Length, Checksum)
 - d. IP Header Analysis (include Source IP, Destination IP, Header Length, TTL, Protocol, Fragmentation(if any))
- Conclusion

References:

1. <https://drive.google.com/file/d/1VVeNtsp4s4KCB9eIJ1lbez1GGylu9xIR/view?usp=sharing>